

CHINMAY MULGUND

+91 8108147801 [✉ chinmaymulgund3559@gmail.com](mailto:chinmaymulgund3559@gmail.com) [in LinkedIn](#) [Github](#)

WORK EXPERIENCE

Dimension Six Technologies Pvt Ltd

Jun 2025 - Present

Embedded Firmware Engineer

Mumbai

- Designed and implemented firmware and PCB layouts for **HOOK UNIQ**, a U-shaped motor-mount attachment that converts a standard bicycle into an e-bike using **ESP-NOW**. Developed a custom wireless throttle with a directional control button, enabling seamless switching between front and rear-wheel drive configurations.
- Developed **HOOK DAILY**, an affordable front-wheel e-conversion solution for delivery personnel, integrating a **Vedder Smart ESC (VESC)** with throttle-based control.
- Built a kiosk-based charging & swap system, including docking hardware, secure device handoff, backend integration, and online payment support.
- Implemented cadence-sensing firmware on an **ISC-nRF52832-A** microcontroller with a **BMA400 accelerometer** to compute pedal RPM and speed, enabling smoother cadence and improved cycling performance.

DeepGrid Semi, Hyderabad

Sep 2024 - Jun 2025

Firmware Engineer Intern

Hyderabad

- Integrated DeepGrid ADAS L0 system with object & lane detection and accident warning for edge deployment on **Raspberry Pi** using the **Hailo8 Accelerator**.
- Developed a system to control a prototype car's front axle using **ESP32** connected to **CARLA** simulation via socket communication, utilizing servo motors for precise control.
- Implemented **BitNetMCU** on **CH32V003 MCU** and **ESP32-S3** for MNIST data prediction, targeting applications like number plate recognition.
- Programmed an **nRF52** development board to measure distance using a custom PCB with **Acconeer A121 Pulsed Coherent Radar Sensor**, developed using **Segger Embedded Studio**.

Paras Telecom Private Limited

Aug 2023 - Nov 2023

Embedded Intern

Pune

- Engineered and optimized **ESP32** firmware for a flexible Wi-Fi setup using a web server, supporting both AP and Station modes with integrated **mDNS** for seamless network discovery.
- Led the design and implementation of a Python-based face recognition access system, achieving over 95% accuracy for secure access control for 100+ users.

Shaurya Racing, VIT Chennai

Oct 2022 - Jan 2025

Electronics & Programming Engineer

Chennai

- Contributed to achieving 7th overall and 5th in statics out of 25+ teams at Formula Bharat 2024.
- Delivered a cutting-edge Gear Position Indicator, enhancing driver awareness and performance by 10% during live testing.
- Reduced data latency by 15% in telemetry systems using **MQTT** for efficient data acquisition and transmission.
- Developed a Driver Information Display, improving real-time data visibility by 20%.

PROJECTS

AI-Powered Bicycle Rear-Collision Alert System: Engineered a computer vision-based warning system utilizing a Raspberry Pi and a Google Coral Accelerator for real-time edge AI vehicle detection, effectively notifying cyclists of rapidly approaching hazards.

Cellular IoT GPS Asset Tracker: Developed a robust real-time tracking architecture using an ESP32 and a SIM A7672S 4G LTE + GNSS Development Board, enabling autonomous telemetry data parsing and transmission over cellular networks.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Embedded C, MATLAB
- **Microcontrollers & SoCs:** ESP32, Arduino UNO/Mega, Xiao ESP32-S3, nRF52 DK, ISC-nRF52832-A, CH32V003, Raspberry Pi
- **Protocols & Interfaces:** ESP-NOW, MQTT, UART, SPI, I2C, Wi-Fi, RF communication, Socket (TCP/IP)
- **Tools & Frameworks:** ESP-IDF, RTOS, Arduino IDE, KiCad, Segger Embedded Studio, VESC & VESC Tool, BitNetMCU, Pygame, NumPy, OpenCV, HEF, ONNX, CARLA
- **Hardware Debugging & Testing:** Digital Oscilloscope, Multimeter, Logic Analyzer, JTAG/SWD Debugging
- **Cloud & Data Analytics:** AWS S3, EC2, Lambda, DynamoDB, AWS Analytics

EDUCATION

Vellore Institute of Technology, Chennai

2021 - 2025

B.Tech in Electronics and Communication Engineering

CGPA: 8.13